

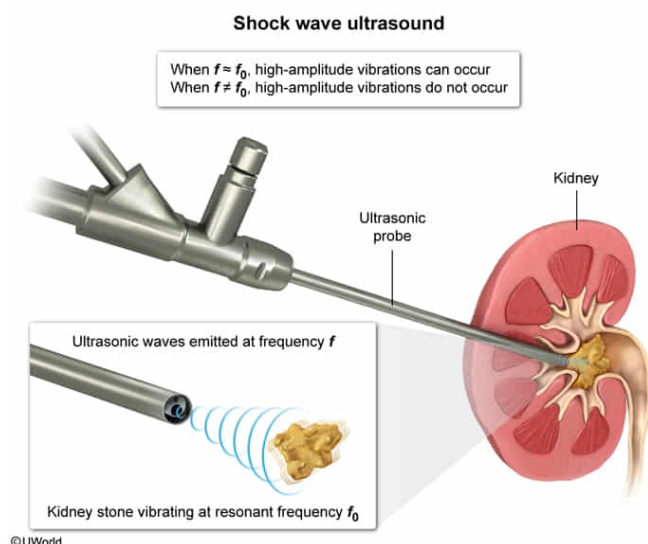
Ultrasonic shock wave devices may be used in medical settings to disrupt disease structures located deep in the body. Among other variables, the device user can control the frequency of emitted waveforms. The adjustment of waveform frequency:

- ☐ A. can be used to provide direct auditory feedback to the user. [5%]
- ✓ ☒ B. enables high-amplitude resonance within diverse target tissues. [44%]
- ☐ C. may eliminate the attenuation effects associated with tissue interfaces. [26%]
- ☐ D. assists in the measurement of bodily fluid velocity. [22%]

Correct

44% answered correctly

Explanation:



Ultrasonic sound waves (ie, ultrasound) are mechanical waves that propagate at a frequency above the upper bound of the human auditory spectrum (~20 kHz). Ultrasonic sound waves behave much like audible **sound waves** that can be detected by the human ear. However, the relatively small wavelength of ultrasonic waves allows them to propagate through body tissues without **diffracting** significantly. Propagation without diffraction enables the use of ultrasound waves in medical imaging contexts.

Unlike ultrasound techniques used for medical imaging, **shock wave ultrasound** is a non-imaging treatment modality in which the mechanical energy of ultrasonic sound waves is transmitted to structures associated with

human disease. Shock wave absorption by these tissues causes **high-amplitude vibrations** that contribute to tissue destruction.

Although **standing wave** vibrations may take place in any object following perturbation, vibrations that occur in non-acoustic objects are generally inaudible because the **amplitude (A)** of the vibrations is too small or because the **resonance frequency (f_0)** falls outside the audible range. The high-amplitude vibrations associated with shock wave ultrasound, for example, occur only because the **shock wave frequency (f)** approaches (or is equal to) the resonance frequency of the target structure.

The resonance frequency of disease structures may change in accordance with size and material composition. Shock wave therapy devices must therefore allow the user to adjust the frequency of the emitted shock waves to ensure effectiveness in destroying particular tissue types.

(Choice A) Shock wave ultrasound utilizes ultrasonic waves that transmit at frequencies beyond the human auditory spectrum, making direct auditory feedback from the device to the user impossible.

(Choice C) Regardless of frequency, all sound waves attenuate (ie, lose intensity) somewhat when **propagating** through a medium. Furthermore, attenuation may occur as sound waves **pass between** media.

(Choice D) **Doppler ultrasound** is a specialized imaging technique that allows for the characterization of dynamic structures. However, shock wave ultrasound is typically used to destroy static structures.

Educational objective:

Shock wave therapies like shock wave ultrasound utilize high-frequency waves to cause destructive, high-amplitude vibrations within target structures. For maximum effectiveness, the frequency of the shock waves should match the resonance frequency of the target structure.

Time spent: 0 sec
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Physics

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Light and Sound